Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students [1]. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people [2].

Introduction courses to fundamental mathematical topics usually look like this: here'e are some basic definitions, let's derive some specific results from them - now go apply these in your field.

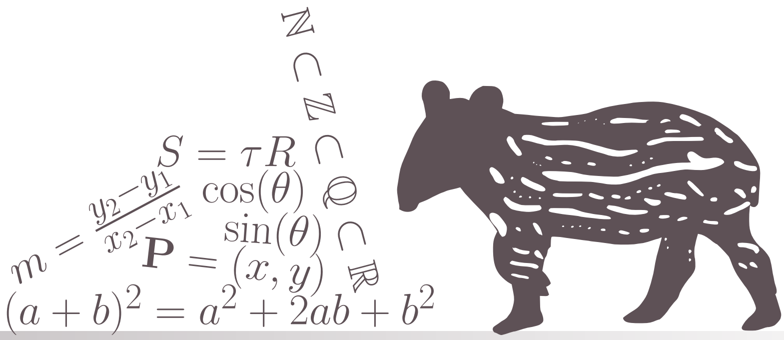
[1] and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

[2] the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

MATHEMATICAL BACKGROUND



$$T(\alpha\vec{x}+\beta\vec{y}) = \alpha T(\vec{x}) + \beta T(\vec{y})$$

$$\langle\vec{a} \mid \vec{b}\rangle = \vec{a}\cdot\vec{b} = \sum_{i=1}^{n} a_i b_i$$

$$\vec{a}\cdot\vec{b} = 0 \Leftrightarrow \vec{a} \perp \vec{b}$$

$$\mathbb{I}_n = \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix}$$

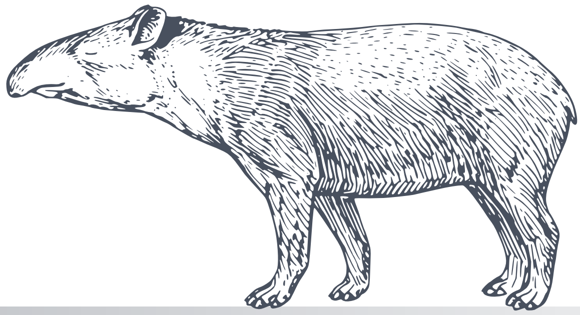
$$(\mathbf{A}\cdot\mathbf{B})^\intercal = \mathbf{B}^\intercal \cdot \mathbf{A}^\intercal$$

$$\vec{v} = \alpha_1\hat{e}_1 + \alpha_2\hat{e}_2 + \dots + \alpha_n\hat{e}_n$$

$$\mathbf{A}\vec{x} = \lambda\vec{x}$$

$$\det(\mathbf{A}\cdot\mathbf{B}) = \det(\mathbf{A})\cdot\det(\mathbf{B})$$

$$\mathbf{C} = \mathbf{A}\cdot\mathbf{B} \Rightarrow c^i_j = \langle a^i \mid b_j\rangle$$



CHAPTER 2

THE BASICS OF LINEAR ALGEBRA

2.1 LINEAR TRANSFORMATIONS

Summary 2.1 The Next Two Sections

This section and the one immediately following it deal with a rather simple, yet very powerful type of functions between vector spaces: **linear transformations**.
In this section we will:

1. see why are we interested in linear transformations,
2. what are their properties, and
3. how they can provide a powerful tool for many different applications.

In the next section we will explore how we can turn the caulculations of linear transformations into an extremely easy and straight-forward process using **matrices**.

2.1.1 Introduction

Text text text...

...

The main purely mathematical property of linear transformations is that they preserve the structure of vector spaces. This means that if we have three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ such that $\vec{c} = \alpha\vec{a} + \beta\vec{b}$, then for any linear transformation T,

$$T(\vec{c}) = \alpha T(\vec{a}) + \beta T(\vec{b}). \tag{2.1}$$

(see Figure 2.1 for a visual example)

However, this is a rather abstract way of discussing linear transformations. Instead, let's first learn some of the more "graphical" properies of linear transformations in the plane, i.e. $T : \mathbb{R}^2 \to \mathbb{R}^2$, and then see practical examples of such transformations before moving on to their more abstract (and powerful) mathematical description.

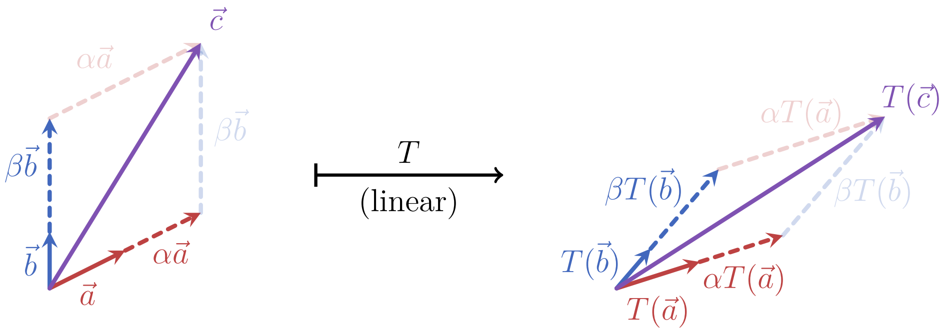


Figure 2.1: Two vectors $\vec{a}$ and $\vec{b}$ are both scaled (by α and β, respectively) and then added together to form the vector $\vec{c}$. All the addition and scaling relations between the vectors are kept under the application of a linear transformation T.

2.1.2 Linear Transformations in 2-Dimensions

(here comes an introductory text about the fundamental linear transformations in $\mathbb{R}^2$, what are their properties and some more relevant information...)

Any linear transformation $T : \mathbb{R}^2 \to \mathbb{R}^2$ always has the following four properties:

- **The origin is kept untouched**: T always maps $\vec{0}$ to $\vec{0}$.

- **Lines remain lines**: a set of points forming a line in $\mathbb{R}^2$ will be mapped to another set of points forming a line (with possibly different direction, distance from origin and scale).

- **Parallel directions remain parallel**: if $\vec{a}$ and $\vec{b}$ point at the same direction, then $\vec{b} = \alpha\vec{a}$ for some $\alpha \in \mathbb{R}$. After the transformation the maps of $\vec{a}$ and $\vec{b}$ will stay parallels, and would even keep their scalar relation: $T(\vec{b}) = \alpha T(\vec{a})$.

- **All areas are scaled by the same factor**: if two shapes S_1 and S_2 have areas A_1 and A_2, respectively, before the transformation is applied, their maps $\tilde{S}_1$ and $\tilde{S}_2$ would have the areas kA_1 and kA_2 (where $k \in \mathbb{R}$ depends on the transformation and can take both positive, zero and negative values).

Figure 2.2 shows these properties in action: there are two planes in the figure - before and after the application of a random linear transformation T (up and down, respectively). The following can be seen:

1. The origin stays at the same place - see how a small purple dot placed on the origin doesn't change its position after the application of T.

2. All lines (e.g. axes, grid lines, the two purple lines) are mapped to lines in the transformed plane.

3. Parallel directions remain parallel, e.g. grid lines aligned with one of the axes, and the two purple lines.

4. All areas are scaled by the same factor: notice for example how all unit squares formed by the grid lines (one is highlighted in orange) are mapped to identical parallelograms. You can also check to see that the blue rectangle is exactly twice as large than the orange square before and after the transformation.

The example in Figure 2.2 also shows what *doesn't* get preserved under a linear transformation. To see this more clearly, Figure 2.3 "zooms-in" on the orange unit square: we label its vertices as $\mathbf{O}, \mathbf{A}, \mathbf{B}$ and $\mathbf{C}$ starting from the point at the origin and going anti-clockwise. The transformed vertices are then, respectively, $\mathbf{O}', \mathbf{A}', \mathbf{B}'$ and $\mathbf{C}'$.

In the original unit square the diagonals are equal: their lengths are $\mathbf{OB} = \mathbf{AC} = \sqrt{2}$. However, it's clear that $\mathbf{O'B'} < \mathbf{A'C'}$. In addition, all the angles in the original unit square are, of course, right angles, but the transformed square is a parallelogram ($\mathbf{O'A'B'C'}$) has two equal acute angles of value α and two equal obtuse angles of value β (where $\beta = \pi - \alpha$).

So we can see that at least two quantities aren't necessarily conserved by linear transformations: (1) **distances**, which unlike areas don't even all scale by the same amount, and (2) **angles** (more generally, **directions**).

One last key observation before we move on: as can be seen in Figure 2.2 and Figure 2.3 linear transfornations in $\mathbb{R}^2$ transform **squares** into **parallelograms**. These parallelograms can be scaled, "squeezed", reflected and/or rotated compared to the original squares.

There are several "basic" linear transformations in $\mathbb{R}^2$: these transformations can be thought of as the building blocks of all possible linear transformations on a 2-dimensional plane. We will see later how we can combine these basic transformations to createnew transformations, but first let's introduce these "basic" linear transformations.

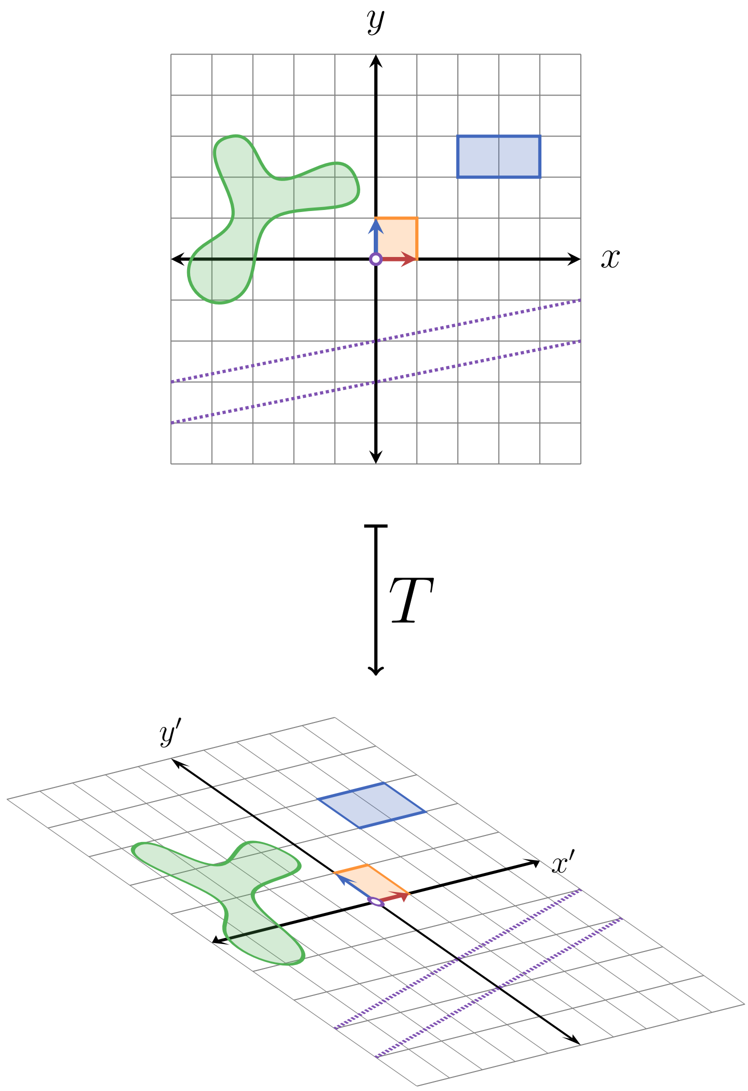


Figure 2.2: Some random linear transformation T applied to $\mathbb{R}^2$, with different shapes showing the four main properties discussed for linear transformations.

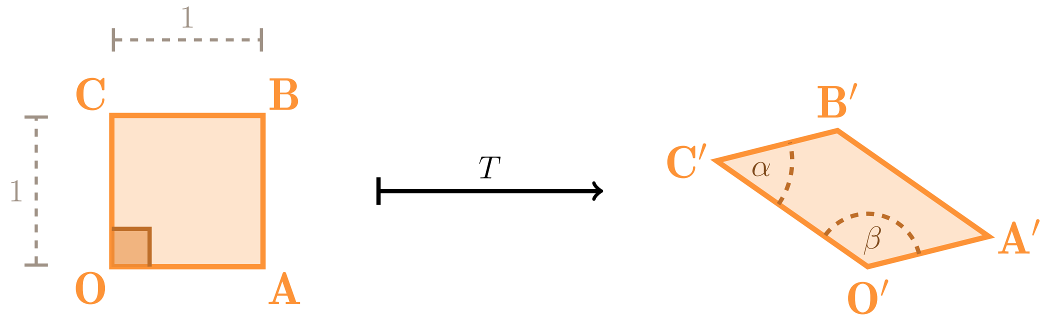


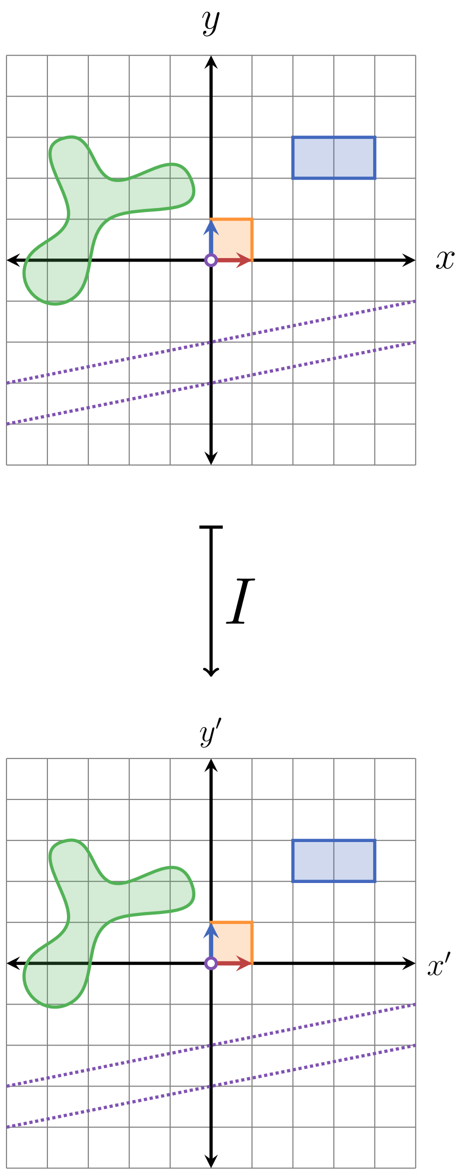
Figure 2.3: Unit square in $\mathbb{R}^2$ before and after the application of the random linear transformation T from Figure 2.2.

In the following introduction, each transformation is shown using a drawing of $\mathbb{R}^2$ with several shapes on it: the two standard basis vectors $\hat{e}_1$ and $\hat{e}_2$, a unit area between the two vectors (drawn in orange), a blue rectangle, two purple parallel lines, a somewhat random green shape[1] and the grid lines (or rather, grid dots). When looking at these figures you should pay attention to how each shape is deformed, and especially what relationship are kept the same or are changed: for example, parallel lines remain parallel - but other angles between lines (including 90°) are not kept. Also note the areas of the different shapes and how they change under the transformation.

Let's begin:

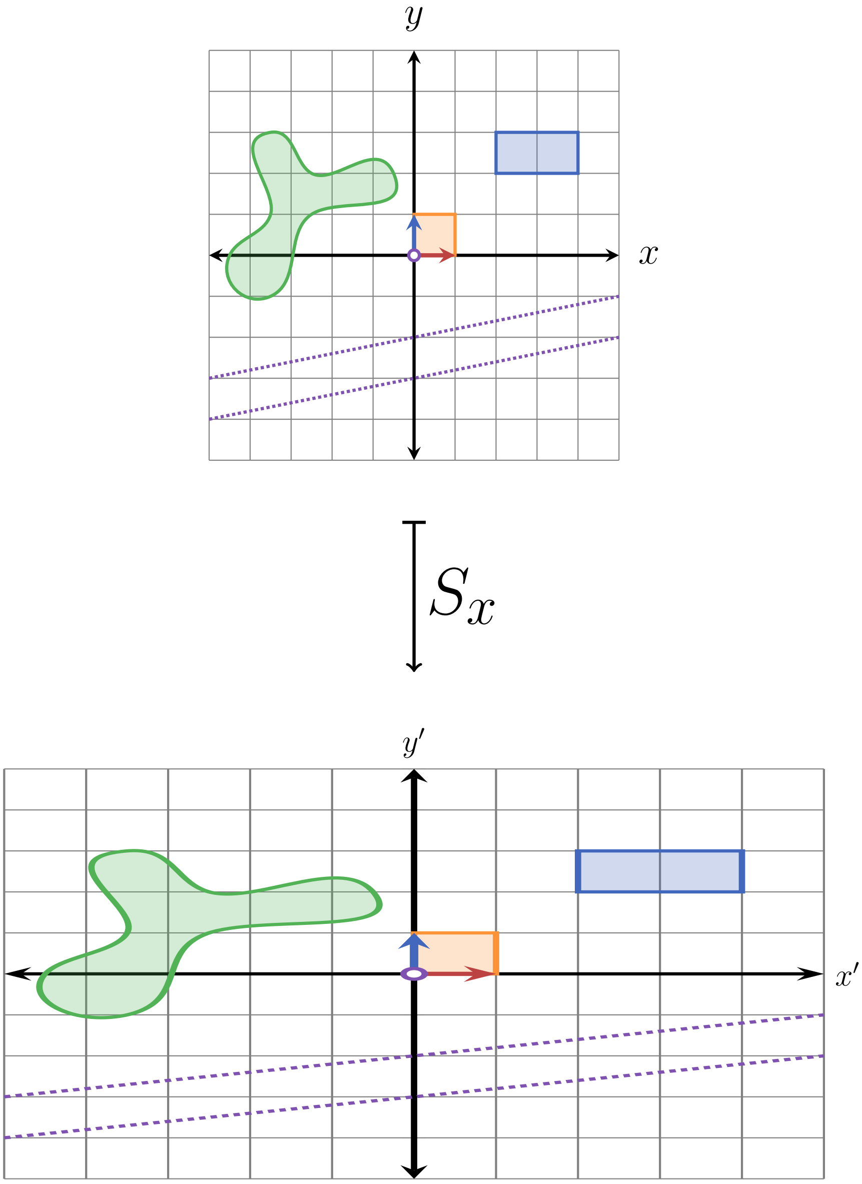
- **The identity transformation**: not doing anything to the plane is a transformation, even if a rather boring one:

[1] I shamelessly stole the shape from https://tikz.org/examples/chapter-12-drawing-smooth-curves/ because it's perfect for this illustration.



Indeed, the identity transformation has all of three key properties of linear transformations: the origin is perserved, parallel lines remain parallel after the transformation, and all areas are scaled by the same factor (namely 1).

- **Scaling**: Scaling the plane can be done in either the x-diretion or the y-direction. Just as with vectors, the scaling factor can be any real number, positive or negative (where negative scaling factors flip the direction of the axis):



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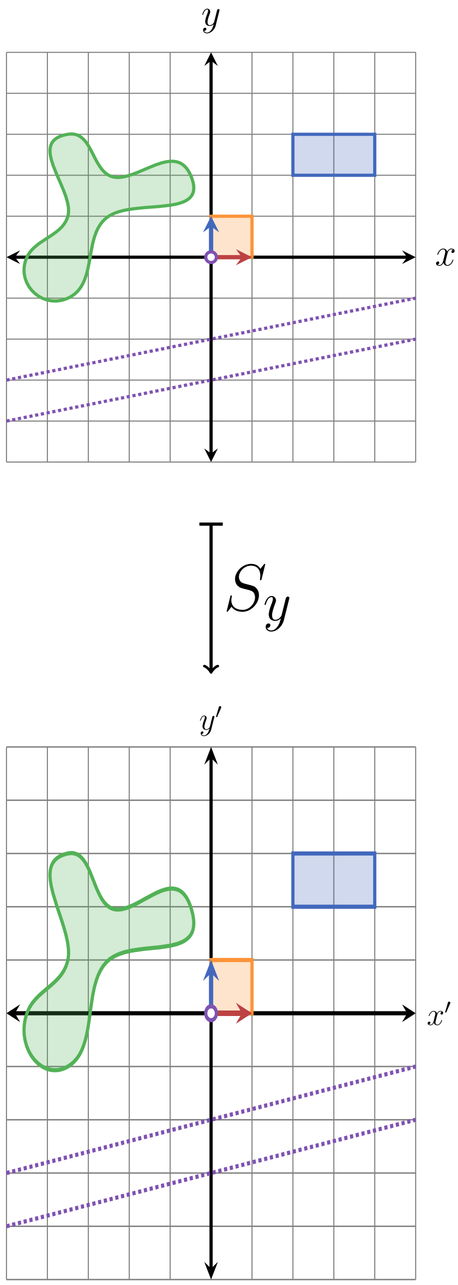
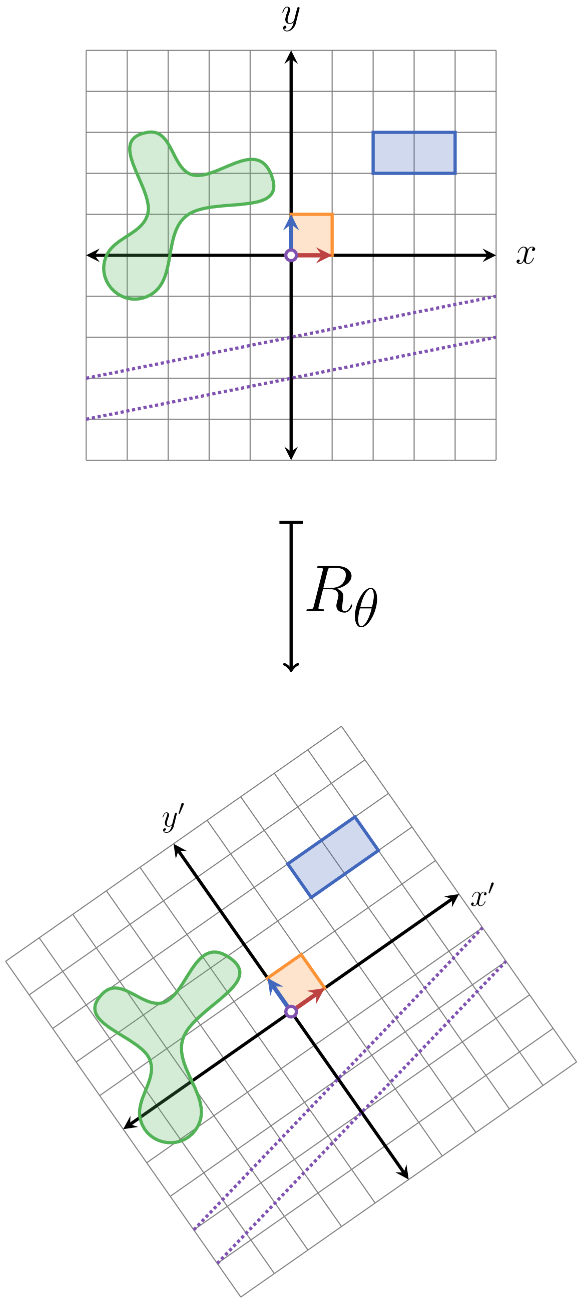
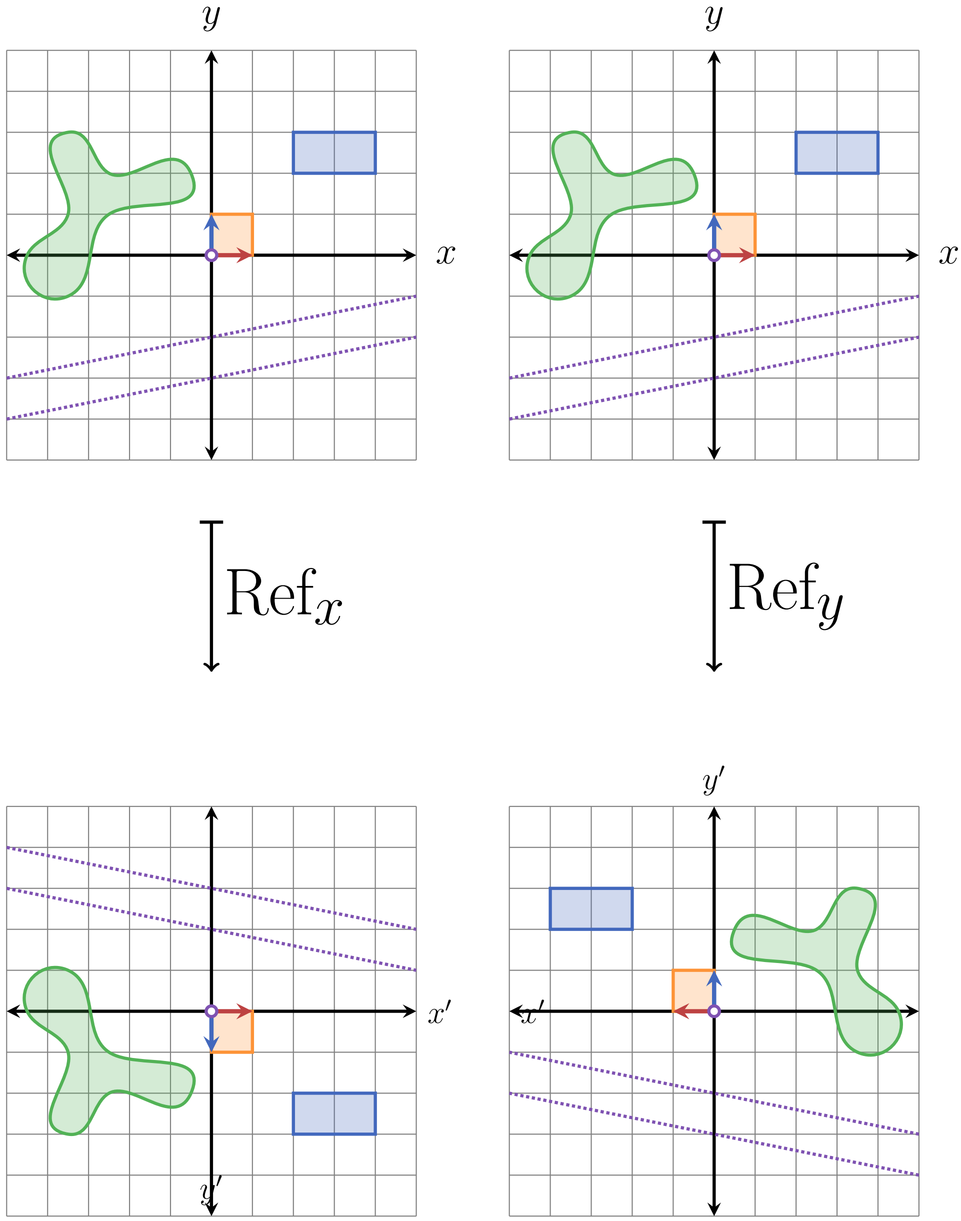


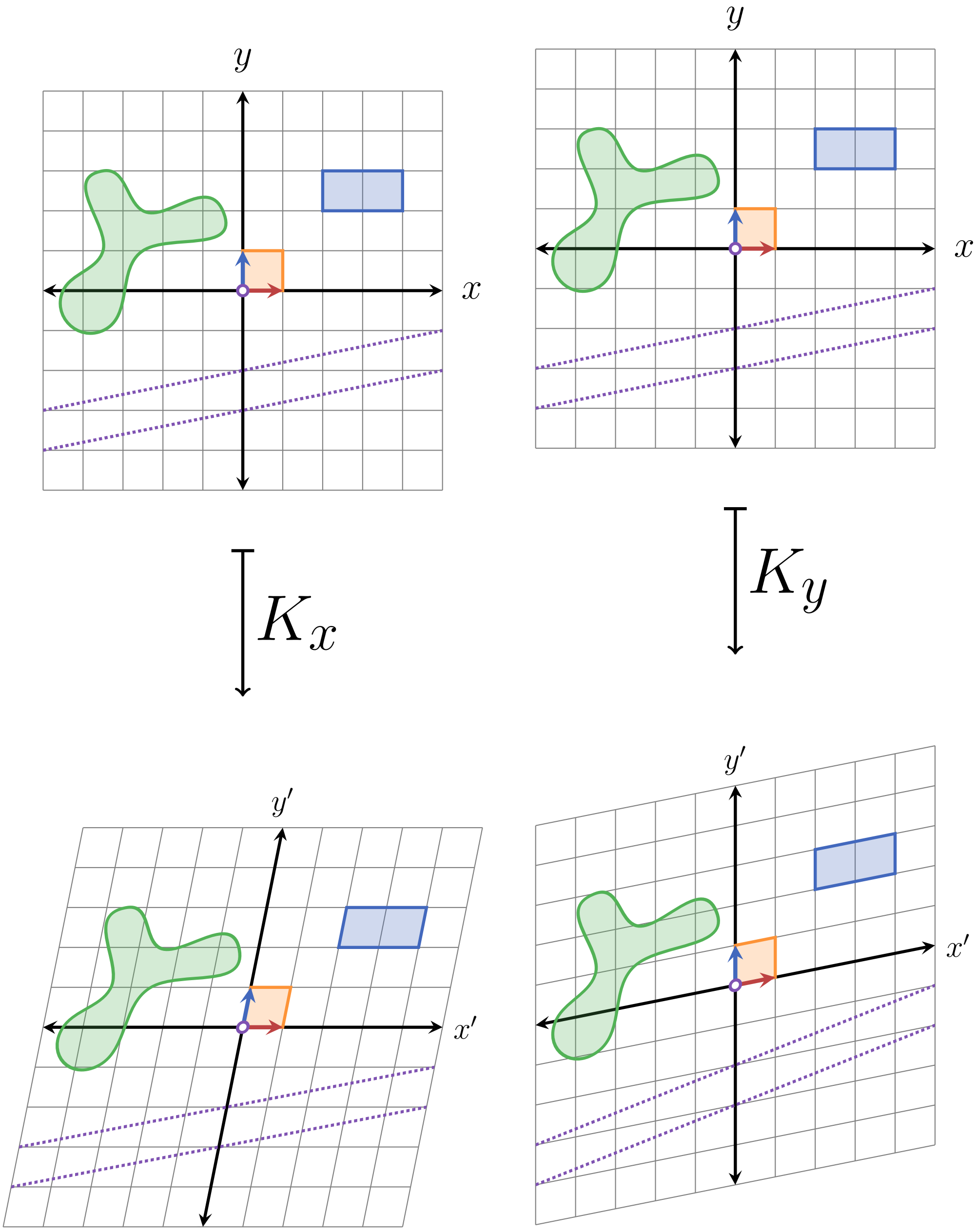
- **Rotation about the origin, anti-clockwise**: text text text:



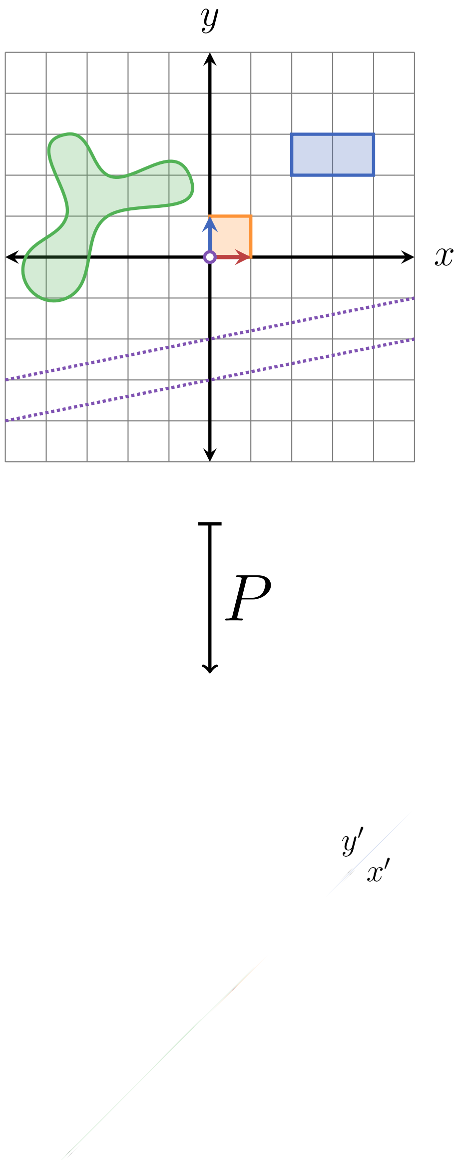
- **Reflection across a line through the origin**: text text text:



- **Shear**: text text text:



- **Projection on line through the origin**: text text text: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2.1.3 Examples of Non-Linear Transformations in 2-Dimensions

2.1.4 3-Dimensional Linear Transformations

2.1.5 Formal Definition

The formal definition of linear transformations between any two vector spaces is the following:

Definition 2.1 Linear Transformation

A transformation $T:\mathbb{R}^n \to \mathbb{R}^m$ is said to be *linear* if for any two vectors $\vec{a},\vec{b}\in\mathbb{R}^n$ and any scalar $\alpha\in\mathbb{R}$:

1. $T\left(\vec{a}+\vec{b}\right)=T(\vec{a})+T(\vec{b})$, and
2. $T\left(\alpha\vec{a}\right)=\alpha T(\vec{a})$.

We can combine the above two criteria using an additional scalar $\beta\in\mathbb{R}$: T is linear if

$$T(\alpha\vec{a}+\beta\vec{b})=\alpha T(\vec{a})+\beta T(\vec{b}).$$

π

Just like with vectors, the first condition is called **additivity** and the second condition is called **homogeneity**. What these conditions say, in simple words, is that we can always scale vectors and add them together before we apply a linear transformation - and the result will be the same as first transforming each vector separately, and then scaling and/or add them together. So in practice we can choose what we want to calculate first (addition and scaling or the transformation) according to what is easier and/or more convinient to calculate.

The formal definition of linear transformations can be used to prove the first three of the four basic geometric properties of linear transformations in 2- and 3-dimensions. In fact, this works for any n-dimensional **Euclidean space** $\mathbb{R}^n$, so let's prove the more general form of these properties instead of limiting ourselves to 2- or 3-dimensions.

Proof 2.1 Geometric Properties of Linear Transformations

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation. Then:

- **The origin is kept untouched**: the origin in $\mathbb{R}^n$ is represented by the n-dimensional zero vector, $\vec{0}$. We can write the zero vector as $0\vec{v}$, where $\vec{v}$ is any vector in $\mathbb{R}^n$ (this is just scaling a vector by 0, which gives the zero vector). When we apply T to $0\vec{v}$ we get

$$T(0\vec{v}) = 0T(\vec{v}) = \vec{0},$$

since $T(\vec{v})$ is a vector in $\mathbb{R}^n$, so scaling it by 0 gives $\vec{0}$.

- **Lines remain lines**: A line in $\mathbb{R}^n$ can be represented just like in 2- and 3-dimensions (see REF TO LINES IN PREVIOUS SECTION):

$$\mathbf{L}(s) = \vec{x}_0 + s\hat{d}.$$

Applying T to $\mathbf{L}$ gives us

$$T\left(\mathbf{L}(s)\right) = T\left(\vec{x}_0 + s\hat{d}\right),$$

and by the linearity of T we can separate the right side of the equality:

$$T\left(\vec{x}_0 + s\hat{d}\right) = T(\vec{x}_0) + sT(\hat{d}).$$

Since both $T(\vec{x}_0)$ and $sT(\hat{d})$ are vectors in $\mathbb{R}^n$, the result of adding them is also a vector in $\mathbb{R}^n$:

$$T(\vec{x}_0) + sT(\hat{d}) = \vec{a} + s\vec{b} = \mathbf{C}(s),$$

where $\mathbf{C}(s)$ is now a function of s with a line structure (vector + variable times a direction) - proving that T transforms lines into lines.

- **Parallel directions remain parallel**: parallel lines have the same direction, i.e. if $\mathbf{L}_1(s)$ and $\mathbf{L}_2(s)$ are parallel, then they can be written as

$$\mathbf{L}_1(s) = \vec{x}_1 + s\hat{d},$$
$$\mathbf{L}_2(s) = \vec{x}_2 + s\hat{d}.$$

(here $\vec{x}_1$ and $\vec{x}_2$ might be different, but $\hat{d}$ - the direction of the lines - is the same for both)

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Applying the linear transformation T on these lines gives

$$\begin{aligned}
T\left(\mathbf{L}_1\right) &= T\left(\vec{x}_1 + s\hat{d}\right) = T\left(\vec{x}_1\right) + sT\left(\hat{d}\right),\\
T\left(\mathbf{L}_2\right) &= T\left(\vec{x}_2 + s\hat{d}\right) = T\left(\vec{x}_2\right) + sT\left(\hat{d}\right),
\end{aligned}$$

and whatever the value of $T(\hat{d})$ is - it is the same direction for both transformed lines. We therefore get that T transforms two parallel lines (i.e. with the same direction) to two parallel lines.

QED

Unfortunately, the last property is rather difficult to prove with the material covered so far, so this proof will have to wait until a later part of the book (REF?).

Let's see how to use the formal definition of linear transformations to actually prove that a given transformation is indeed linear. We can start with the identity transformation since it's a very simple transformation.

- **Additivity**: for any $\vec{a}, \vec{b} \in \mathbb{R}^2$, we have

$$I(\vec{a}) = \vec{a},\ \ I(\vec{b}) = \vec{b}, \tag{2.2}$$

and also that

$$I\left(\vec{a} + \vec{b}\right) = \vec{a} + \vec{b}. \tag{2.3}$$

Therefore, if we first add the vectors $\vec{a}$ and $\vec{b}$ together and then apply I on the result, we get

$$I\left(\vec{a} + \vec{b}\right) \overset{\text{from Equation 2.3}}{=} \vec{a} + \vec{b} \underset{\text{from Equation 2.2}}{=} I(\vec{a}) + I(\vec{b}). \tag{2.4}$$

- **Homogeneity**: for any vector $\vec{a} \in \mathbb{R}^2$ and scalar $\alpha \in \mathbb{R}$ we have

$$I\left(\alpha\vec{a}\right) = \alpha\vec{a}, \tag{2.5}$$

and together with Equation 2.2 (specifically its first equality) we get

$$I\left(\alpha\vec{a}\right) = \alpha\vec{a} = \alpha I(\vec{a}). \tag{2.6}$$

Fullfilling these two conditions proves that I is indeed a linear transformation.

Next, let's see that rotation (about the origin) is also a linear transformation, starting with the simple 90° anti-clockwise rotation. Recall that a 2-dimensional vector $\vec{a}=\begin{bmatrix}a_x\\a_y\end{bmatrix}$ can be rotated by 90° anti-clockwise by exchanging its components and then adding a minus sign to the first component (REF TO TRICK):

$$\begin{bmatrix}a_x\\a_y\end{bmatrix}\xmapsto[\text{(anti-clockwise)}]{90^\circ\ \text{rotation}}\begin{bmatrix}-a_y\\a_x\end{bmatrix}.$$

Using this fact we can easily show that a 90° rotation is a linear transformation:

- **Additivity**: we need two vectors, let's call them

$$\vec{a}=\begin{bmatrix}a_x\\a_y\end{bmatrix}\text{ and }\vec{b}=\begin{bmatrix}b_x\\b_y\end{bmatrix}.$$

Then, adding the vectors first and then rotating the result we get

$$\mathrm{Rot}_{90^\circ}\left(\vec{a}+\vec{b}\right)=\mathrm{Rot}_{90^\circ}\left(\begin{bmatrix}a_x\\a_y\end{bmatrix}+\begin{bmatrix}b_x\\b_y\end{bmatrix}\right)=\mathrm{Rot}_{90^\circ}\left(\begin{bmatrix}a_x+b_x\\a_y+b_y\end{bmatrix}\right)=\begin{bmatrix}-\left(a_y+b_y\right)\\a_x+b_x\end{bmatrix}.$$

And if we first rotate the vectors and then add the result we get

$$\mathrm{Rot}_{90^\circ}\left(\vec{a}\right)+\mathrm{Rot}_{90^\circ}\left(\vec{b}\right)=\begin{bmatrix}-a_y\\a_x\end{bmatrix}+\begin{bmatrix}-b_y\\b_x\end{bmatrix}=\begin{bmatrix}-(a_y+b_y)\\a_x+b_x\end{bmatrix},$$

exactly the same as before.

- **Homogeneity**: if we scale $\vec{a}$ by some scalar $\lambda\in\mathbb{R}$ first and then rotating the result we get

$$\mathrm{Rot}_{90^\circ}\left(\lambda\vec{a}\right)=\mathrm{Rot}_{90^\circ}\left(\lambda\begin{bmatrix}-a_y\\a_x\end{bmatrix}\right)=\mathrm{Rot}_{90^\circ}\left(\begin{bmatrix}\lambda a_x\\\lambda a_y\end{bmatrix}\right)=\begin{bmatrix}-\lambda a_y\\\lambda a_x\end{bmatrix}.$$

And if we first rotate $\vec{a}$ and then scale it by λ we get

$$\lambda\mathrm{Rot}_{90^\circ}\left(\vec{a}\right)=\lambda\begin{bmatrix}-a_y\\a_x\end{bmatrix}=\begin{bmatrix}-\lambda a_y\\\lambda a_x\end{bmatrix},$$

again - the same result as before.

Together this means that rotating around the origin anti-clockwise by 90° is indeed a linear transformation.

Moving on, let's show that a rotation around the origin by *any* angle θ is a linear transformation. To start, we need to know how to transform a general 2-dimensional vector $\vec{a} = \begin{bmatrix} a_x \\ a_y \end{bmatrix}$, which we will find, of course, by using basic trigonometry (use Figure 2.4 to follow these steps). First, we calculate the norm of $\vec{a}$ (which we'll call $\mathbf{a}$ for convenience):

$$\mathbf{a} = \|\vec{a}\| = \sqrt{a_x^2 + a_y^2}. \tag{2.7}$$

We can then express $\vec{a}$ in polar coordinates (REF), as it's a more natural way of dealing with angles (and thus - rotations):

$$a_x = \mathbf{a}\cos(\alpha), \tag{2.8}$$
$$a_y = \mathbf{a}\sin(\alpha). \tag{2.9}$$

Rotating the $\vec{a}$ by θ yields a new vector $\vec{b}$ which as the same norm as $\vec{a}$ (since rotating a vector doesn't change its norm), and an angle $\beta = \alpha + \theta$. Therefore in polar coordinates $\vec{b}$ is

$$b_x = \mathbf{a}\cos(\beta) = \mathbf{a}\cos(\alpha + \theta), \tag{2.10}$$
$$b_y = \mathbf{a}\sin(\beta) = \mathbf{a}\sin(\alpha + \theta). \tag{2.11}$$

Using the trigonomitric identities for the *cos* and *sin* of a sum of two angles (REF), we get:

$$b_x = \mathbf{a}\cos(\alpha + \theta) = \mathbf{a}\left[\cos(\alpha)\cos(\theta) - \sin(\alpha)\sin(\theta)\right], \tag{2.12}$$
$$b_y = \mathbf{a}\sin(\alpha + \theta) = \mathbf{a}\left[\sin(\alpha)\cos(\theta) + \sin(\alpha)\cos(\theta)\right]. \tag{2.13}$$
$$\tag{2.14}$$

Now, we can express α in terms of the inverse trigonometric functions arccos and arcsin:

$$\alpha = \arccos\left(\frac{a_x}{\mathbf{a}}\right) = \arcsin\left(\frac{a_y}{\mathbf{a}}\right). \tag{2.15}$$

Substituting Equation 2.15 into Equation 2.14 we get

$$\begin{aligned}
b_x &= \mathbf{a}\left[\cos(\alpha)\cos(\theta) - \sin(\alpha)\sin(\theta)\right] \\
&= \mathbf{a}\left[\cos\left(\arccos\left(\frac{a_x}{\mathbf{a}}\right)\right)\cos(\theta) - \sin\left(\arcsin\left(\frac{a_y}{\mathbf{a}}\right)\right)\sin(\theta)\right] \\
&= \cancel{\mathbf{a}}\left[\frac{a_x}{\cancel{\mathbf{a}}}\cos(\theta) - \frac{a_y}{\cancel{\mathbf{a}}}\sin(\theta)\right] \\
&= a_x\cos(\theta) - a_y\sin(\theta),
\end{aligned} \tag{2.16}$$

$$\begin{aligned}
b_y &= \mathbf{a}\left[\sin(\alpha)\cos(\theta) - \sin(\alpha)\cos(\theta)\right] \\
&= \mathbf{a}\left[\sin\left(\arcsin\left(\frac{a_y}{\mathbf{a}}\right)\right)\cos(\theta) - \sin\left(\arcsin\left(\frac{a_y}{\mathbf{a}}\right)\right)\cos(\theta)\right] \\
&= \cancel{\mathbf{a}}\left[\frac{a_y}{\cancel{\mathbf{a}}}\cos(\theta) - \frac{a_x}{\cancel{\mathbf{a}}}\sin(\theta)\right] \\
&= a_x\sin(\theta) + a_y\cos(\theta).
\end{aligned} \tag{2.17}$$

To sum up this result, here it is again - this time in “mapping” notation:

$$\begin{bmatrix} a_x \\ a_y \end{bmatrix} \xmapsto{\mathrm{R}_\theta} \begin{bmatrix} a_x\cos(\theta) - a_y\sin(\theta) \\ a_x\sin(\theta) + a_y\cos(\theta) \end{bmatrix}. \tag{2.18}$$

Now that we finally have a formula for rotating any vector in $\mathbb{R}^2$ by any angle about the origin, we can finally prove that this is indeed a linear transformation.

- **Additivity**: let

$$\vec{a} = \begin{bmatrix} a_x \\ a_y \end{bmatrix},\ \vec{b} = \begin{bmatrix} b_x \\ b_y \end{bmatrix}.$$

Adding the two together gives

$$\vec{a} + \vec{b} = \begin{bmatrix} a_x + b_x \\ a_y + b_y \end{bmatrix},$$

and then rotating the result by θ yields

$$\mathrm{Rot}_\theta\left(\vec{a} + \vec{b}\right) = \begin{bmatrix} (a_x + b_x)\cos(\theta) - \left(a_y + b_y\right)\sin(\theta) \\ (a_x + b_x)\sin(\theta) + \left(a_y + b_y\right)\cos(\theta) \end{bmatrix}.$$

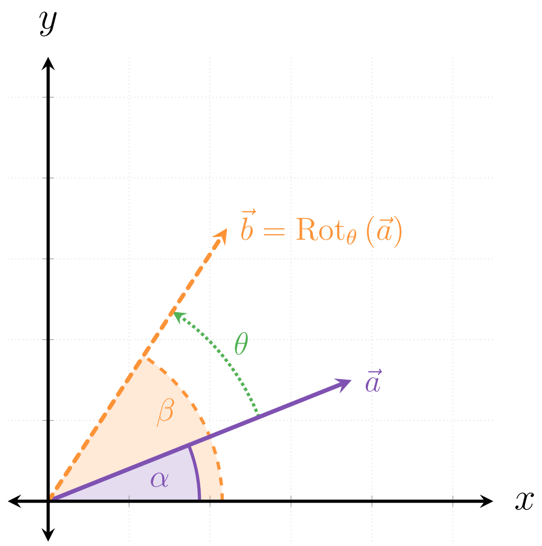

Figure 2.4: The vector $\vec{a}$ is rotated by an angle θ anti-clockwise about the origin, yielding the vector $\vec{b}$. $\vec{a}$ is at θ relative to the x-axis, which becomes $\beta = \alpha + \theta$ after the rotation.

On the other hand, rotating the two vectors first gives

$$\mathrm{Rot}_\theta(\vec{a}) = \begin{bmatrix} a_x\cos(\theta) - a_y\sin(\theta) \\ a_x\sin(\theta) + a_y\cos(\theta) \end{bmatrix},$$
$$\mathrm{Rot}_\theta\left(\vec{b}\right) = \begin{bmatrix} b_x\cos(\theta) - b_y\sin(\theta) \\ b_x\sin(\theta) + b_y\cos(\theta) \end{bmatrix}.$$

Finally, adding the two rotated vectors together we get

$$\begin{aligned}\mathrm{Rot}_\theta(\vec{a}) + \mathrm{Rot}_\theta\left(\vec{b}\right) &= \begin{bmatrix} a_x\cos(\theta) - a_y\sin(\theta) \\ a_x\sin(\theta) + a_y\cos(\theta) \end{bmatrix} + \begin{bmatrix} b_x\cos(\theta) - b_y\sin(\theta) \\ b_x\sin(\theta) + b_y\cos(\theta) \end{bmatrix}, \\ &= \begin{bmatrix} (a_x + b_x)\cos(\theta) - \left(a_y + b_y\right)\sin(\theta) \\ (a_x + b_x)\sin(\theta) + \left(a_y + b_y\right)\cos(\theta) \end{bmatrix}.\end{aligned}$$

- **Homogeneity**: Scaling $\vec{a}$ by a scalar $s \in \mathbb{R}$ and then rotating it by θ gives

$$\mathrm{Rot}_\theta(s\vec{a}) = \mathrm{R}_\theta \begin{bmatrix} sa_x \\ sa_y \end{bmatrix} = \begin{bmatrix} sa_x\cos(\theta) - sa_y\sin(\theta) \\ sa_x\sin(\theta) + sa_y\cos(\theta) \end{bmatrix}.$$

On the other hand, first rotating $\vec{a}$ and the scaling by s gives

$$s\mathrm{Rot}_{\theta}\left(\vec{a}\right) = s\begin{bmatrix} a_x\cos(\theta) - a_y\sin(\theta) \\ a_x\sin(\theta) + a_y\cos(\theta) \end{bmatrix} = \begin{bmatrix} sa_x\cos(\theta) - sa_y\sin(\theta) \\ sa_x\sin(\theta) + sa_y\cos(\theta) \end{bmatrix}.$$

Hoorah! Rotation is indeed a linear transformation.

2.1.6 Composing Linear Transformations

In the discussion about 2- and 3-dimensional linear transformations I mentioned several times compositions of such transformations and simply stated that they are also linear transformations. This topic, however, deserves some more exploration. First, in order for us to even be able to compose two transformations T_1 and T_2, the composition needs to be properly defined - specifically, if we want to compose T_2 onto T_1 (that is, first apply T_1 and then apply T_2 on the result), the **domain** of T_1 must be at least a superset of the **image** of T_1. Otherwise, we might get a situation where we apply T_1 to some vector and then not being able to apply T_2 on the result, since it isn't defined.

TBW: (1) more text about the domain and image of functions in relation to composition, (2) discussion on how the order of composition matters (i.e. it isn't commutative).

...

Proof 2.2 The Composition of Linear Transformations is Linear

Let $T_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $T_2 : \mathbb{R}^m \to \mathbb{R}^k$ be linear transformations, let $\vec{v}, \vec{w} \in \mathbb{R}^n$ and let $\lambda \in \mathbb{R}$. Then:

- **Additivity**:

$$\begin{aligned} \left(T_1 \circ T_2\right)\left(\vec{v} + \vec{w}\right) &= T_2\left(T_1\left(\vec{v} + \vec{w}\right)\right) \\ &= T_2\left(T_1(\vec{v}) + T_1(\vec{w})\right) \\ &= T_2\left(T_1(\vec{v})\right) + T_2\left(T_1(\vec{w})\right) \\ &= \left(T_1 \circ T_2\right)(\vec{v}) + \left(T_1 \circ T_2\right)(\vec{w}). \end{aligned}$$

- **Homogeneity**:

$$\left(T_1 \circ T_2\right)\left(\lambda\vec{v}\right) = T_2\left(T_1\left(\lambda\vec{v}\right)\right)$$

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$$
\begin{aligned}
&= T_2\left(\lambda T_1\left(\vec{v}\right)\right) \\
&= \lambda\Big(T_2\left(T_1(\vec{v})\right)\Big) \\
&= \lambda\left(T_1 \circ T_2\right)\left(\vec{v}\right).
\end{aligned}
$$

Therefore, $(T_1 \circ T_2)$ is also a linear transformation.

QED

2.1.7 General Forms of Linear Transformations

You might have noticed by now that all of the linear transformations we dealt with had a similar structure, involving adding and scaling the different components of the vectors they act on. Indeed, the most general form of a linear transform in $\mathbb{R}^2$ is the following:

$$
T\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix}, \tag{2.19}
$$

where a, b, c and d are some real numbers. In a sense, this is just two linear combinations of the vector's components: one for the x-component and one for the y-component. Performing any other operation would make the transformation no longer linear. To see this, let's first quickly prove that this general form is indeed a linear transformation:

Proof 2.3 General 2-Dimensional Linear Transformation

Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ a transformation defined for $\vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix} \in \mathbb{R}^2$ as

$$
T\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix},
$$

with $a, b, c, d \in \mathbb{R}$. As usual, we will prove that T confirms to the two criteria of linear transformations.

- **Additivity**: Let $\vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ and $\vec{w} = \begin{bmatrix} w_x \\ w_y \end{bmatrix}$. Then first adding the two

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vectors together and then applying T we get

$$T\left(\vec{v}+\vec{w}\right) = T\begin{bmatrix} v_x + w_x \\ v_y + w_y \end{bmatrix} = \begin{bmatrix} a(v_x+w_x)+b(v_y+w_y) \\ c(v_x+w_x)+d(v_y+w_y) \end{bmatrix}.$$

And on the other hand, transforming each vector separately first and then adding the results, we get

$$\begin{aligned} T(\vec{v})+T(\vec{w}) &= \begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix} + \begin{bmatrix} aw_x + bw_y \\ cw_x + dw_y \end{bmatrix} \\ &= \begin{bmatrix} av_x + bv_y + aw_x + bw_y \\ cv_x + dv_y + cw_x + dw_y \end{bmatrix} \\ &= \begin{bmatrix} a(v_x+w_x)+b(v_y+w_y) \\ c(v_x+w_x)+d(v_y+w_y) \end{bmatrix}. \end{aligned}$$

- **Homogeneity**: Let $\vec{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$ and $\lambda \in \mathbb{R}$. Then scaling $\vec{v}$ by λ and then applying T gives us

$$T\left(\lambda\vec{v}\right) = T\begin{bmatrix} \lambda v_x \\ \lambda v_y \end{bmatrix} = \begin{bmatrix} a\lambda v_x + b\lambda v_y \\ c\lambda v_x + d\lambda v_y \end{bmatrix} = \lambda\begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix}.$$

And applying T to $\vec{v}$ first and then scaling it by λ gives

$$\lambda T(\vec{v}) = \lambda T\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \lambda\begin{bmatrix} av_x + bv_y \\ cv_x + dv_y \end{bmatrix}.$$

Therefore, T is a linear transformation.

QED

2.1.8 Calculating Linear Transformations, Part 1

Linear transformations have a very nice property which makes applying them on different vectors an extremely efficient process. Consider the linear transformation $T:\mathbb{R}^2 \to \mathbb{R}^2$ which is a composition of scaling by a factor of 2 in the y-diretion followed by a rotation of 90° anti-clockwise around the origin. Now let's say that we wish to calculate how T transforms the vector

$$\vec{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

In principle, we could simply apply the operations one after the other on $\vec{a}$:

$$\vec{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \xmapsto{\times 2 \text{ in } y} \begin{bmatrix} 3 \\ 4 \end{bmatrix} \xmapsto{90^\circ \text{ rotation}} \begin{bmatrix} -4 \\ 3 \end{bmatrix} = T(\vec{a}). \tag{2.20}$$

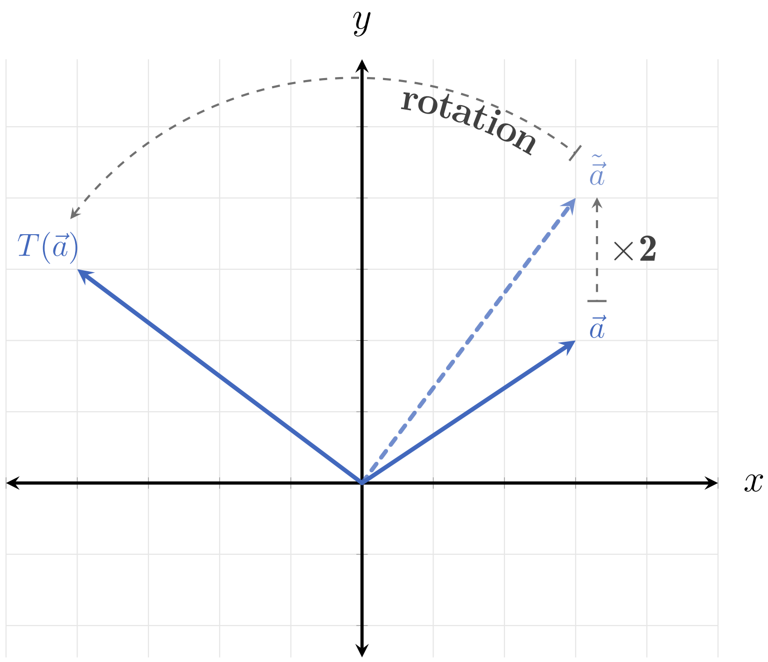

Figure 2.5: The transformation T applied to $\vec{a}$: first the vector is scaled by 2 in the y-direction, and then rotated by 90° anti-clockwise.

Now, what happens if we wish to calculate a more complicated linear transformation, which is both composed out of many more basic transformations - and also acts on higher-dimensional vectors? Performing each transformation step might not be so straight-forward, and is definitely not an efficient process. In fact, even if we're dealing with a simple example just like the one we just saw, we would still need to apply it every time we want to calculate how a new vector is transformed.

It turns out that thanks to the simplicity of linear transformations [2] we only need to perform a few simple calculations per transformation to practically get any possible application of it on any vector!

To see how this is the case, let's go back to our simple 2-dimensional example. First, recall that we can express any vector in terms of the standard basis vectors (REF), which in this case gives us (note the color-coding)

$$\vec{a} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 3\overset{\substack{\hat{e}_1 \\ \downarrow}}{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} + 2\overset{\substack{\hat{e}_2 \\ \downarrow}}{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}. \tag{2.21}$$

We can then apply the defining properties of linear transformations (REF) on $\vec{a}$:

$$\begin{aligned} T\begin{bmatrix} 3 \\ 2 \end{bmatrix} &= T\left(3\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= T\left(3\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) + T\left(2\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) \\ &= 3T\begin{bmatrix} 1 \\ 0 \end{bmatrix} + 2T\begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ &= 3T\left(\hat{e}_1\right) + 2T\left(\hat{e}_2\right). \end{aligned} \tag{2.22}$$

Now, applying T to the two standard basis vectors $\hat{e}_1$ and $\hat{e}_2$ we get

$$\begin{aligned} \hat{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} &\xmapsto[\times 2 \text{ in } y]{} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \xmapsto[90^\circ \text{ rotation}]{} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = T(\hat{e}_1) \\ \hat{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} &\longmapsto \begin{bmatrix} 0 \\ 2 \end{bmatrix} \longmapsto \begin{bmatrix} -2 \\ 0 \end{bmatrix} = T(\hat{e}_2) \end{aligned} \tag{2.23}$$

And when we use these transformed basis vectors in the linear combination we

[2] more correctly: to the strict rules they enforce

finally get

$$\begin{aligned} 3T\left(\hat{e}_1\right) + 2T\left(\hat{e}_2\right) &= 3\begin{bmatrix} 0 \\ 1 \end{bmatrix} + 2\begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 3 \end{bmatrix} + \begin{bmatrix} -4 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} -4 \\ 3 \end{bmatrix}, \end{aligned} \tag{2.24}$$

which is exactly what we got when we calculated T's effect directly on $\vec{a}$ (cf. Equation 2.26).

Notice exactly what happened: in the standard basis representation, $\vec{a}$ had the coefficients 3 and 2 for $\hat{e}_1$ and $\hat{e}_2$, respectively. When applying T to $\vec{a}$ these two coefficients remained, but the basis changed: instead of $\left\{\hat{e}_1, \hat{e}_2\right\}$, the two basis vectors became $\left\{T(\hat{e}_1), T(\hat{e}_2)\right\}$.

Of course, this is not unique to $\vec{a}$. For example, let's look at another vector,

$$\vec{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix}.$$

Its coefficients in the standard basis representation are 2 and -4, respectively. So we expect $T(\vec{b})$ to be

$$\begin{aligned} T(\vec{b}) &= 2T(\hat{e}_1) - 4T(\hat{e}_2) \\ &= 2\begin{bmatrix} 0 \\ 1 \end{bmatrix} - 4\begin{bmatrix} -2 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 2 \end{bmatrix} + \begin{bmatrix} 8 \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 2 \end{bmatrix}. \end{aligned} \tag{2.25}$$

If we now calculate $T(\vec{b})$ directly, we get the following:

$$\vec{b} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} \xmapsto{\times 2 \text{ in } y} \begin{bmatrix} 2 \\ -8 \end{bmatrix} \xmapsto{90^\circ \text{ rotation}} \begin{bmatrix} 8 \\ 2 \end{bmatrix} = T(\vec{b}). \tag{2.26}$$

What this insight shows us is that given a linear transformation, we only need to know how it transforms the respective standard basis vectors in order to calculate how it transform any vector in its domain. The transformed version of the vector would simply be the linear combination of these transformed basis vectors, with the coefficients given by the vector's presentation in the standard basis. It really can't get any simpler than that!

To make this a bit more formal, let's look at the most generic case: say $T : \mathbb{R}^n \to \mathbb{R}^n$ is a linear transformation [3]. Then applying T to a vector $\vec{v} \in \mathbb{R}^n$ in its standard basis representation,

$$\vec{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

we get

$$\begin{aligned} T(\vec{v}) &= T \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= T\left(v_1 \hat{e}_1 + v_2 \hat{e}_2 + \cdots + v_n \hat{e}_n \right) \\ &= v_1 T(\hat{e}_1) + v_2 T(\hat{e}_2) + \cdots + v_n T(\hat{e}_n). \end{aligned} \tag{2.27}$$

This is an incredibly powerful insight which allows us to save a lot of calculations - and which we will use in the next section to make the calculations of linear transformations *even more* concise and simple, using the power of...

Actual page layout values.

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[3] we'll deal with the more general case of $T : \mathbb{R}^n \to \mathbb{R}^m$ where $n \neq m$ in the next section.